Assignment Title

Designing and Evaluating Software Testing Strategies for a Real-World System

Assignment Objectives

This assignment is designed to help students:

- Understand the importance of software testing in quality assurance (LO1)
- Identify and apply different testing techniques (LO2)
- Design effective test cases using appropriate methods (LO3)
- Analyze risks and apply test management concepts (LO4)
- Explore testing tools and compare their usage (LO5)

Assignment Scenario

Select **ONE** of the following systems (or get approval for another system):

- Online Shopping System
- Library Management System
- Banking System
- Hospital Management System
- Student Management System

You are working as a **Software Quality Assurance Engineer** responsible for ensuring the quality of the selected system.

Tasks

Task 1: Introduction to Software Testing

- Define **software testing**
- Explain the **importance of testing in software quality assurance**
- Describe **at least 3 reasons for software failures**

Task 2: Testing Levels & Types

- Explain the following testing levels:
    - Unit Testing
    - Integration Testing
    - System Testing
    - Acceptance Testing
- Identify and explain **2 testing types** relevant to your system (e.g., performance, security)

Task 3: Test Case Design

- Select **one module** of your system (e.g., login, payment, registration)
- Design **at least 8–10 test cases** using:
    - Equivalence Partitioning OR
    - Boundary Value Analysis

Include:

- Test Case ID
- Description
- Input Data
- Expected Output
- Actual Output (assume results)
- Status (Pass/Fail)

Task 4: White Box vs Black Box Testing

- Explain:
    - White Box Testing
    - Black Box Testing
- Compare both methods (minimum 4 differences)
- State which is more suitable for your selected module and why

Task 5: Risk Analysis

- Identify **at least 4 risks** in your system:
    - Product Risks
    - Project Risks
- Create a simple **Risk Matrix** (Likelihood vs Impact)

Task 6: Test Automation Tools

- Select **one testing tool** (e.g., Selenium, JUnit, TestNG)
- Describe:
    - Features
    - Advantages
    - Limitations
- Explain how it can be used in your selected system

Task 7: Conclusion

- Summarize your findings
- Reflect on the importance of QA in software development